IntelliAIDrive: Traffic Sign Classifier + RL Policy

Graciella Mhervie D. Jimenez
Holy Angel University
School of Computing
Angeles City, Pampanga
+63-929-856-9503
gdjimenez@student.hau.edu.ph

Jian Kalel D. Marquez
Holy Angel University
School of Computing
Angeles City, Pampanga
+63-962-642-0273
jdmarquez@student.hau.edu.ph

Arron Kian M. Parejas
Holy Angel University
School of Computing
Angeles City, Pampanga
+63-969-137-9979
amparejas@student.hau.edu.ph

Jenica Sarah B. Tongol
Holy Angel University
School of Computing
Angeles City, Pampanga
+63-962-119-6098
jbtongol2@student.hau.edu.ph

ABSTRACT

Autonomous driving systems require both accurate visual perception and reliable sequential decision-making. However, directly feeding raw image pixels into a reinforcement learning policy often introduces high-dimensional noise, increases computational cost, and weakens policy stability. This study presents IntelliAIDrive, a hybrid framework that combines lightweight visual classification with reinforcement learning for simulated autonomous navigation. The system uses YOLOv8n classification to recognize traffic signs from a multi-class traffic sign dataset, then integrates the resulting label information into a reinforcement learning environment for decision-making. The model was trained and evaluated using a Kaggle traffic sign dataset containing 58 classes, 4,170 training images, and 1,994 test images. Experimental results showed that the final model achieved 95.08% accuracy, 93.38% F1-score, 92.08% precision, and 95.08% recall. These results suggest that decoupling visual recognition from policy learning can improve performance and produce a more stable and interpretable autonomous navigation pipeline.

CCS Concepts

• Computing methodologies → Computer vision • Computing methodologies → Reinforcement learning • Computing methodologies → Neural networks

Keywords

Autonomous Driving, Traffic Sign Classification, YOLOv8, Reinforcement Learning, PPO, Computer Vision

1. INTRODUCTION

Autonomous driving remains one of the most challenging applications of artificial intelligence because it requires a system to perceive its environment, understand road rules, and make safe decisions in real time [1]. One important part of this problem is traffic sign recognition, since a vehicle must correctly interpret signs such as speed limits, stop signs, and directional restrictions before it can respond appropriately [5]. If visual perception fails, the entire decision-making pipeline becomes unreliable.

Recent studies have shown that deep learning methods significantly improve traffic sign recognition performance, especially when lightweight models are used for efficient inference [2], [5]. At the same time, reinforcement learning has gained attention for its ability to learn sequential actions in dynamic environments, making it useful for autonomous navigation tasks [3], [4]. These ideas support the use of modular autonomous driving pipelines that separate perception from control rather than forcing one model to solve both problems at once.

This paper presents IntelliAIDrive, a hybrid framework that combines traffic sign classification and reinforcement learning in a simulated setting. Instead of using raw image pixels directly as the state representation for the policy, the system first performs traffic sign classification and then uses the resulting structured information inside the decision-making environment. This separation reduces input complexity while maintaining strong predictive performance.

2. PROBLEM STATEMENT

Although many computer vision models can detect or classify traffic signs with high accuracy, recognition alone is not sufficient for autonomous navigation [4], [5]. A self-driving system must also decide how to act after interpreting a sign. For example, recognizing a stop sign is only useful if the navigation policy can correctly translate that recognition into an appropriate action.

A common difficulty in reinforcement learning for visual environments is that raw pixel input is large, noisy, and computationally expensive [3]. This can lead to unstable learning, inefficient policy updates, and poor convergence. Because of this, there is a need for a more structured approach in which perception and decision-making are separated into clearer components.

Therefore, this study aims to design and evaluate a hybrid autonomous navigation framework that combines a lightweight traffic sign classification model with a reinforcement learning policy. The goal is to improve classification performance, reduce unnecessary input complexity, and support more stable navigation behavior in simulation.

3. RELATED LITERATURE REVIEW

3.1 Traffic Sign Recognition

Traffic sign recognition is a core requirement in autonomous driving because road signs communicate legal restrictions, warnings, and required behaviors [1], [5]. The German Traffic Sign Recognition Benchmark established one of the most widely used benchmark settings for multi-class traffic sign classification and helped standardize evaluation in this area [1]. More recent literature also shows that traffic sign recognition systems remain sensitive to real-world issues such as occlusion, lighting variation, motion blur, and domain shift [5].

YOLO-based systems are often associated with detection, but the Ultralytics implementation also supports image classification, which makes it suitable for efficient whole-image traffic sign categorization when the task setup is classification rather than localization [2]. In academic prototypes, this lightweight classification approach can be practical because it reduces computational overhead while preserving strong recognition performance.

3.2 Reinforcement Learning for Navigation

Reinforcement learning has been widely used for sequential decision-making because it enables agents to learn actions by interacting with an environment and maximizing long-term reward [3]. In autonomous navigation, this is valuable because the agent must continuously choose actions based on current observations and expected future outcomes [4]. Policy-based methods such as Proximal Policy Optimization are especially useful because they support stable updates and have become a strong baseline for continuous and discrete control tasks [3].

Recent reviews of RL in autonomous driving also emphasize that observation design and reward shaping strongly affect performance [4]. If the state representation is noisy or overly complex, the policy may struggle to learn effectively. This supports the idea of pairing reinforcement learning with a separate perception module instead of training directly from raw image input.

3.3 Human-Centered and Language-Aware Systems

Human-centered intelligent systems increasingly incorporate language-aware interfaces to improve interpretability and user interaction. In the context of this study, language-based modules are not the core evaluated contribution, but they motivate a possible future extension in which higher-level user commands influence behavior without overriding fixed safety constraints. This direction is consistent with broader trends in making intelligent transportation systems more explainable and controllable.

4. METHODOLOGY

IntelliAIDrive follows a hybrid pipeline composed of dataset preparation, visual classification, and reinforcement learning-based decision-making. The main goal of this design is to reduce the complexity of the policy input by separating perception from action selection.

4.1 Dataset and Preprocessing

The study used a Kaggle traffic sign classification dataset containing 58 traffic sign classes, 4,170 training images, and 1,994 test images. All images were organized by class and prepared for model training. During preprocessing, the dataset was standardized for use with the selected classification architecture. The image size used during training was 64 × 64 pixels, which allowed faster processing while preserving relevant visual details.

Table 1. Dataset Summary

Dataset Attribute	Value
Number of Classes	58
Total Training Images	4,170
Training Split	3,313
Validation Split	857
Test Images	1,994
Image Size	64 × 64 pixels
Train-Validation Split Ratio	80:20

In addition to resizing the images to 64 × 64 pixels, the dataset was organized to preserve class-level separation between training, validation, and testing. This setup helped reduce leakage between splits and ensured that evaluation results better reflected the model's ability to generalize to unseen samples. Because traffic sign datasets often contain visually similar categories, maintaining a clean split was important for measuring whether the classifier could distinguish between related classes under a controlled experimental setting.

4.2 Traffic Sign Classification Using YOLOv8n

For the perception stage, the system used YOLOv8n classification as a lightweight traffic sign classifier. The model was initialized from a pretrained checkpoint and fine-tuned on the traffic sign dataset. The training process used 15 epochs, a batch size of 64, and image resolution of 64 × 64.

The reason for selecting YOLOv8n was its lightweight structure and efficient inference characteristics, which are useful in time-sensitive intelligent systems [2]. Although YOLO is often associated with detection, in this project the classification variant was used specifically for multi-class traffic sign recognition.

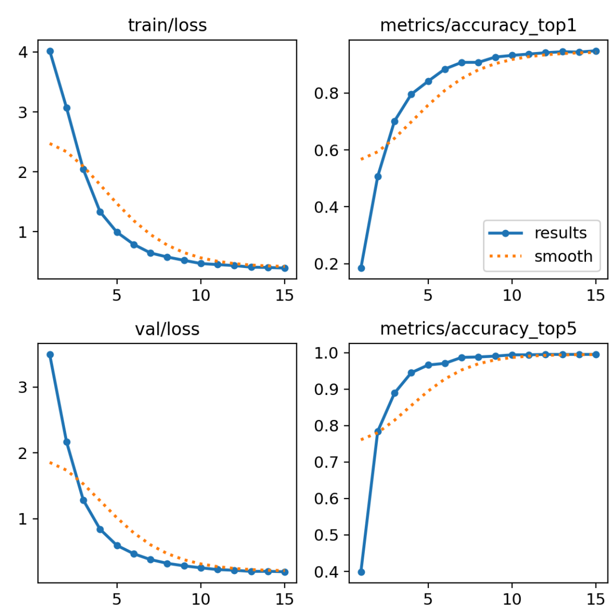

Figure 1. YOLOv8n classification training results showing training loss, validation loss, top-1 accuracy, and top-5 accuracy across 15 epochs.

4.3 Reinforcement Learning Environment

After classification, the project transitioned into a reinforcement learning stage in which the recognized traffic sign information was used within a custom navigation environment. The environment was implemented using Gymnasium, and the learning algorithm was based on PPO through the Stable-Baselines3 framework.

The policy was trained using a learning rate of 0.0003, discount factor $\gamma = 0.99$, and clip range of 0.2. These settings were chosen to encourage stable policy updates and support long-term reward learning. The reinforcement learning environment simulated decision-making based on traffic-related inputs rather than relying only on direct image pixels.

The reinforcement learning stage was designed to evaluate whether structured visual information could support downstream decision-making more effectively than raw image input alone. By passing traffic sign recognition outputs into the environment, the policy was able to operate on a simpler and more interpretable state representation. This reduced the burden on the agent and made the training setup more suitable for an academic prototype focused on controlled experimentation rather than full real-world driving complexity.

4.4 Evaluation Metrics

The final model was evaluated using standard classification metrics, namely accuracy, precision, recall, and F1-score. A confusion matrix was also generated to examine class-level prediction performance. In addition, training and evaluation plots were used to visualize the behavior of the model and identify strengths or weaknesses across classes.

5. EXPERIMENTAL SETUP

The experiments were conducted in a Python-based notebook environment using common machine learning libraries such as Ultralytics, PyTorch, Gymnasium, Stable-Baselines3, NumPy, Pandas, Matplotlib, and Scikit-learn. The pipeline consisted of two major stages: (1) YOLOv8n classification training and (2) reinforcement learning policy evaluation.

For the classification stage, the dataset was split into training and validation sets using an approximate 80:20 ratio, resulting in 3,313 training images and 857 validation images. The test set remained separate for final evaluation. For the reinforcement learning stage, the agent interacted with a custom environment that incorporated sign-based observations and reward-driven navigation decisions.

Table 2. Hyperparameter Settings

Parameter	Value
Model	YOLOv8n-cls
Number of Classes	58
Image Size	64 × 64
Epochs	15
Batch Size	64
PPO Learning Rate	0.0003
Gamma	0.99
Clip Range	0.2

6. RESULTS AND DISCUSSION

The final evaluation results show that IntelliAIDrive performed strongly on the traffic sign classification task. The model achieved 95.08% accuracy, 93.38% F1-score, 92.08% precision, and 95.08% recall during final evaluation. These scores indicate that the model was able to correctly classify most traffic sign categories while maintaining balanced predictive performance across the dataset.

Table 4. Final Performance Metrics

Metric	Value
Accuracy	95.08%
Precision	92.08%
Recall	95.08%
F1-Score	93.38%

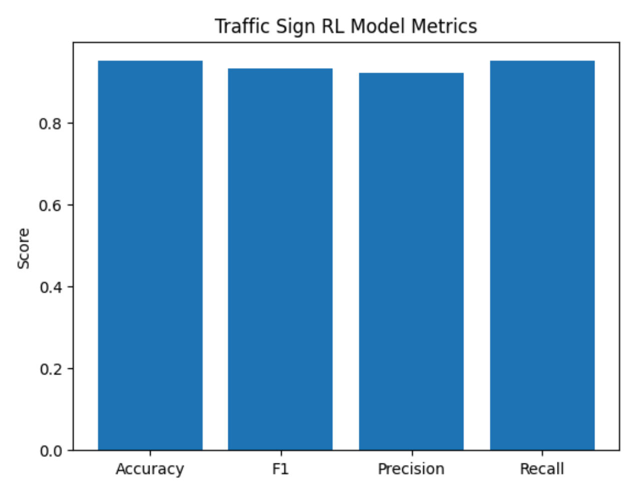

Figure 2. Final evaluation metrics of the IntelliAIDrive model, including accuracy, F1-score, precision, and recall.

As shown in Figure 2, the final model achieved strong and balanced performance across all major evaluation metrics. The model reached 95.08% accuracy, 93.38% F1-score, 92.08% precision, and 95.08% recall, indicating reliable traffic sign classification performance across the evaluation set.

Beyond the high overall scores, the training plots also suggest that the classification model converged in a stable manner. As shown in Figure 1, both training loss and validation loss decreased steadily across the 15 training epochs, while top-1 and top-5 accuracy improved consistently. This trend indicates that the model learned useful class-discriminative features without showing obvious instability during optimization. In the context of this study, stable classification performance was important because the reinforcement learning component depended on reliable sign recognition outputs for downstream decision support.

The confusion matrix also showed that most predictions were concentrated along the diagonal, which means the majority of samples were classified correctly. Misclassifications were present in a smaller number of classes, which is expected in a multi-class traffic sign task where some sign categories may share similar visual features.

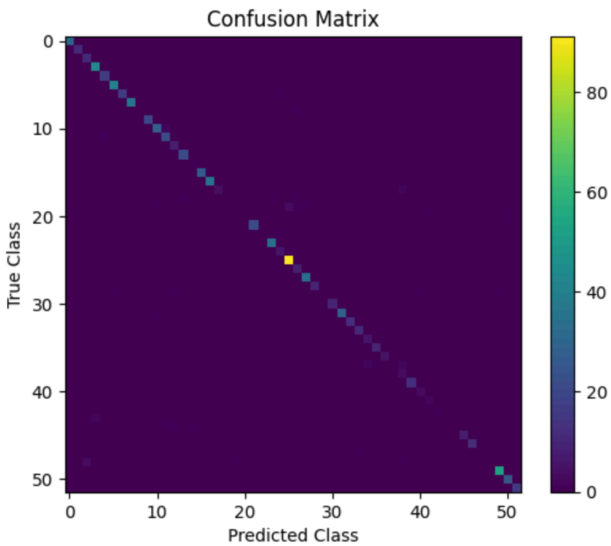

Figure 3. Confusion matrix showing class-level prediction performance of the final IntelliAIDrive traffic sign classifier.

Additional evaluation plots further support the classifier's behavior at the class level. The precision-recall curves for the top five classes show strong average precision values and suggest that the model was highly effective at identifying these selected categories with minimal false positives and false negatives. This does not mean that all classes performed identically, but it does indicate that at least the strongest categories were learned with high confidence.

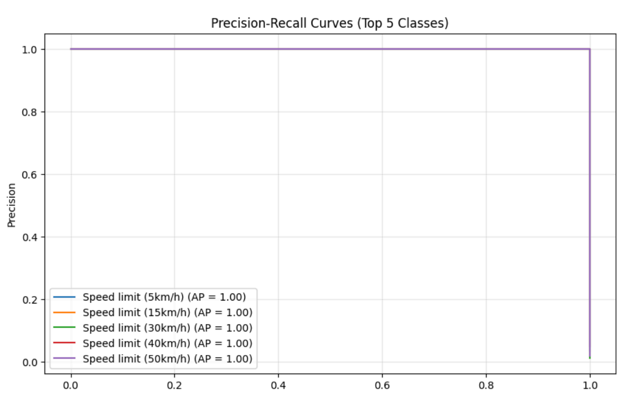

Figure 4. Precision-recall curves for the top five traffic sign classes, showing strong class-wise performance with average precision values close to 1.00.

Overall, the results suggest that the hybrid design was effective in reducing the burden on the decision-making stage. Instead of requiring the reinforcement learning component to learn directly from raw image pixels, the framework first extracted structured sign information through classification. This made the overall system more manageable, interpretable, and suitable for controlled experimentation. This observation is further supported by the ablation summary in Table 3, where weaker configurations were associated with noisier observations and less reliable downstream support.

6.1 Design Trade-Off Analysis

To better understand the contribution of the final system design, the project analyzed the structured sign-based pipeline against weaker alternatives considered during development. These alternatives included less structured observations for the decision stage and earlier classifier configurations with lower expected maturity. The comparison was used as an ablation-style analysis to examine whether the final hybrid setup provided clearer, more stable, and more interpretable downstream behavior.

Table 3. Ablation and Design Trade-Off Summary

Configuration	Description	Expected / Observed Effect
Final Hybrid Pipeline	YOLOv8n classification + sign-based RL input	Best overall clarity and stability
Weaker Input Representation	Less structured observations to decision stage	Higher noise, lower interpretability
Reduced Classification Maturity	Earlier or weaker classifier setup	Less reliable downstream support

6.2 Error and Slice Analysis

Error analysis was conducted using the confusion matrix and selected precision-recall curves. Most predictions were concentrated along the diagonal of the confusion matrix, indicating that the model correctly classified the majority of traffic sign samples. However, a limited number of off-diagonal errors were still observed, suggesting confusion between visually similar categories.

The precision-recall curves for the top five classes showed near-perfect average precision values, but these results should be interpreted carefully because they represent only a subset of the full class set. Taken together, the error analysis suggests that the classifier is strong overall but still sensitive to class similarity, which remains an important limitation for traffic sign recognition tasks.

7. ETHICAL CONSIDERATIONS AND LIMITATIONS

This study was conducted strictly as an academic and simulation-based project. The proposed framework is not intended for deployment in real-world vehicles. While the results are promising, the system has only been evaluated on a benchmark-style traffic sign dataset and in a custom reinforcement learning environment. As noted in the literature, real-world deployment introduces additional challenges such as domain shift, weather variation, changing illumination, sensor noise, and unexpected object behavior [4], [5].

Another limitation is that high performance on classification metrics does not guarantee fully safe driving behavior in complex real traffic settings. Autonomous systems require much broader perception, planning, and safety validation than what is covered in this project. The current work should therefore be interpreted as a proof-of-concept framework for combining lightweight vision and reinforcement learning in a controlled environment.

Another important limitation of the study is that the reported results were obtained under controlled data and simulation conditions. The traffic sign classes used in evaluation do not fully represent the visual variability of real roads, where signs may be damaged, partially blocked, poorly illuminated, or captured at difficult angles. In the same way, the reinforcement learning environment simplifies many factors that affect real driving, such as pedestrians, multi-vehicle interactions, road texture, weather, and sensor uncertainty. Because of these constraints, the results should be interpreted as evidence of technical feasibility rather than proof of real-world readiness.

8. REPRODUCIBILITY AND IMPLEMENTATION NOTES

To support reproducibility, the project was organized with explicit environment and execution files, including requirements.txt, run.sh, and supporting project documentation. Training and evaluation were performed in a Python-based environment using Ultralytics, PyTorch, Stable-Baselines3, Gymnasium, NumPy, Pandas, Matplotlib, and Scikit-learn. The final report, proposal, checkpoint, and supporting documentation were maintained alongside the repository to preserve consistency between implementation and written analysis.

9. CONCLUSION

This paper presented IntelliAIDrive, a hybrid framework that combines traffic sign classification and reinforcement learning for simulated autonomous navigation. By separating visual recognition from policy learning, the system reduced input complexity and achieved strong predictive performance. Using a traffic sign dataset with 58 classes, the final model achieved 95.08% accuracy, 93.38% F1-score, 92.08% precision, and 95.08% recall.

The findings suggest that modular autonomous driving pipelines remain valuable, especially in academic environments where interpretability, efficiency, and manageable experimentation are important. Future improvements to IntelliAIDrive may focus on strengthening both perception and decision-making under more realistic conditions. On the perception side, future work may include broader class coverage, stronger augmentation, and testing under conditions such as glare, blur, and occlusion. On the decision-making side, the reinforcement learning environment may be expanded to include denser traffic flow, more dynamic obstacles, and more complex action constraints. The framework may also be extended toward human-in-the-loop or multi-agent settings to better reflect the challenges of intelligent transportation research [3], [4].

10. ACKNOWLEDGMENTS

The authors acknowledge their shared contributions in the areas of integration, modeling, evaluation, data governance, and documentation.

11. REFERENCES

[1] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel. 2011. The German Traffic Sign Recognition Benchmark: A multi-class classification competition. 2011 International Joint Conference on Neural Networks.

[2] Ultralytics. Image Classification. Ultralytics Documentation.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. 2017. Proximal Policy Optimization Algorithms. arXiv preprint arXiv:1707.06347.

[4] A. Irshayyid, A. Alqaralleh, M. S. Almalawi, and M. M. Althobaiti. 2024. A review on reinforcement learning-based highway autonomous driving merge decision making. AI and Engineering Applications 2 (2024), 100050.

[5] H. Chen, M. A. H. Ali, Y. Nukman, B. A. Razak, S. Turaev, Y. Chen, S. Zhang, Z. Huang, Z. Wang, and R. Abdulghafor. 2024. Computational methods for automatic traffic signs recognition in autonomous driving on road: A systematic review. Results in Engineering 24 (2024), 103553.